Lecture 16: Big data and machine learning I

PPHA 34600

Prof. Fiona Burlig

Harris School of Public Policy
University of Chicago

From last time: fuzzy regression discontinuity

As usual, we'd like to run:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

The regression discontinuity:

- Suppose D_i is determined by whether or not X_i lies above a cutoff, c
- **Idea**: Having X_i just above or just below c is as good as random...
- ... And there is a discontinuous change in D_i as a result of crossing c

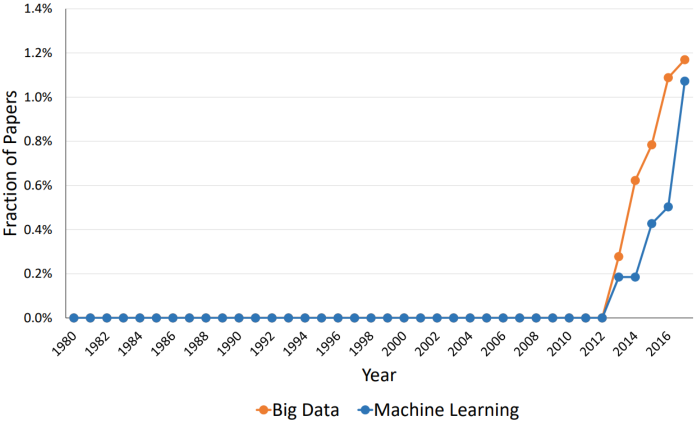
→ We can compare Y_i for units with X_i just above c to Y_i for units with X_i just below c

- With incomplete changes in D_i from $X_i < c$ to $X_i \geq c$:

$$D_i = \alpha + \gamma \mathbf{1}[X_i \geq c] + f(X_i) + \varepsilon_i \text{ for } c - h < X_i < c + h$$

$$Y_i = \alpha + \tau \hat{D}_i + f(X_i) + \varepsilon_i \text{ for } c - h < X_i < c + h$$

And now for something completely different...



How big is Big?

A computer scientist might say:

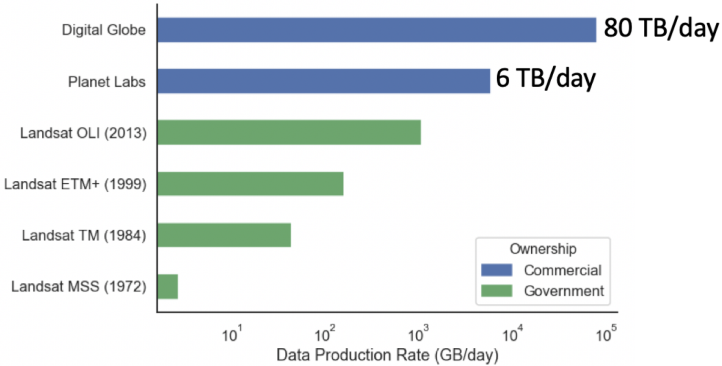
- "Too big to fit on your computer"

An economist might say:

- "I dunno, 15 GB?"

$\rightarrow$ All of this is a bit fuzzy

Big data are getting bigger every day

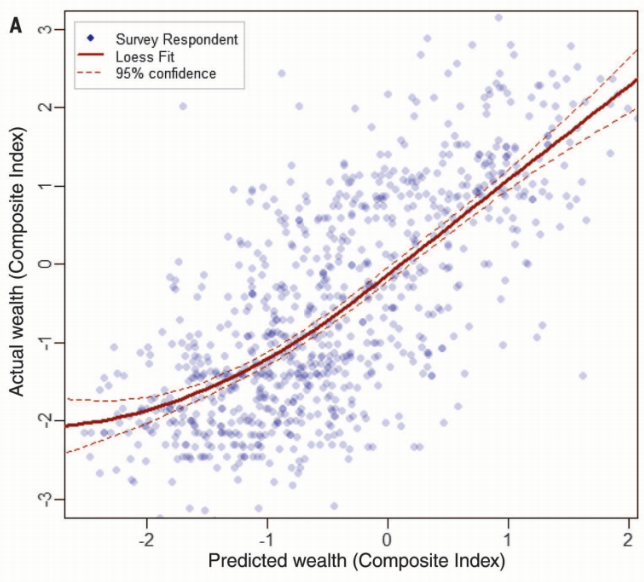


Thinking outside of the box to get new data

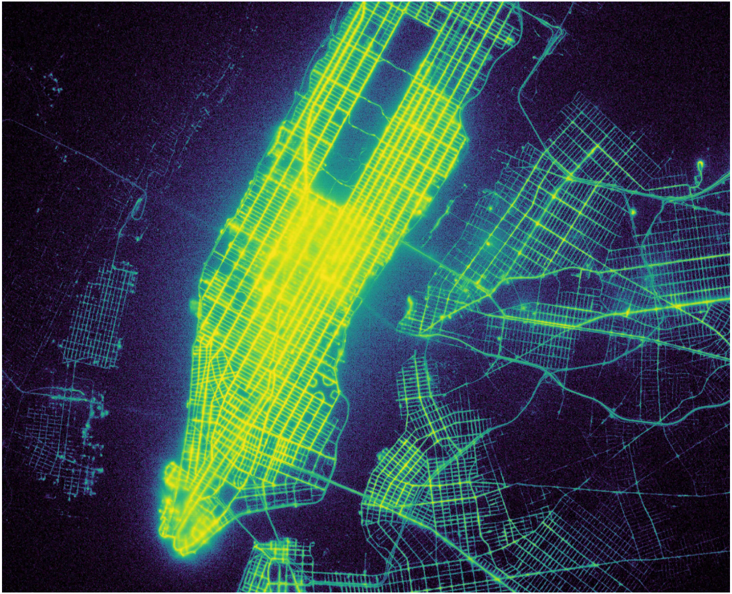
The era of Big Data isn't just good because of small standard errors:

- New data collection methods present opportunities
- We can study previously unanswerable questions
- Sometimes this requires getting a bit creative

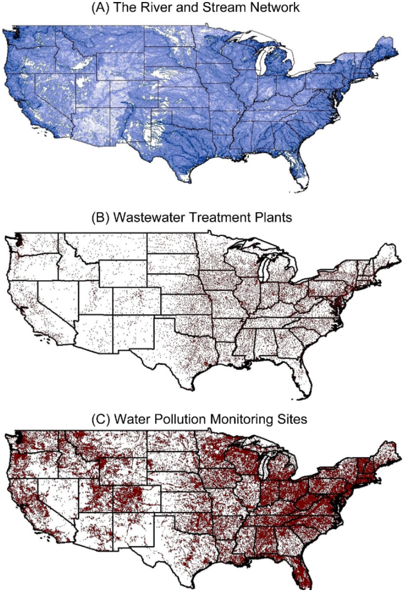
Thinking outside of the box to get new data



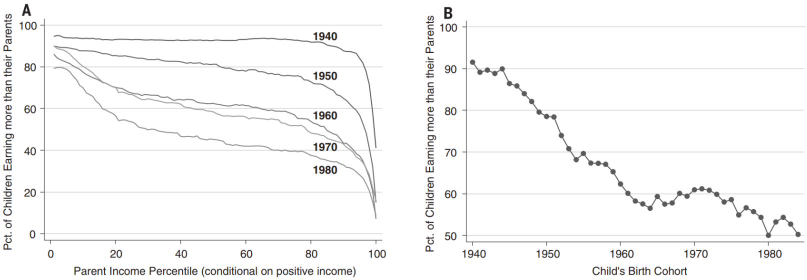
Thinking outside of the box to get new data



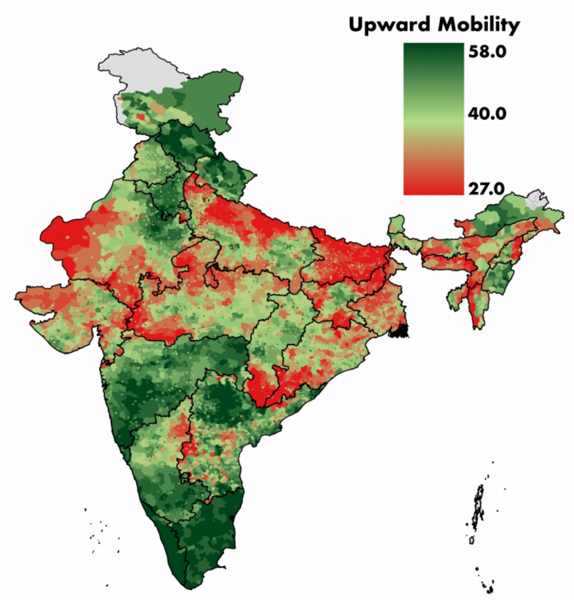
Thinking outside of the box to get new data



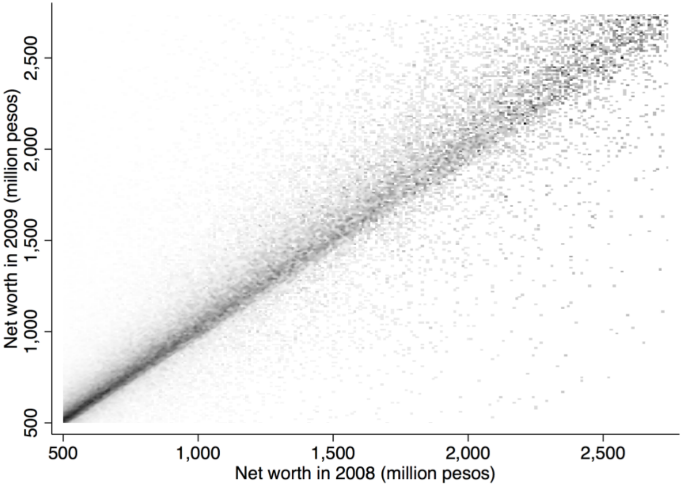
Thinking outside of the box to get new data



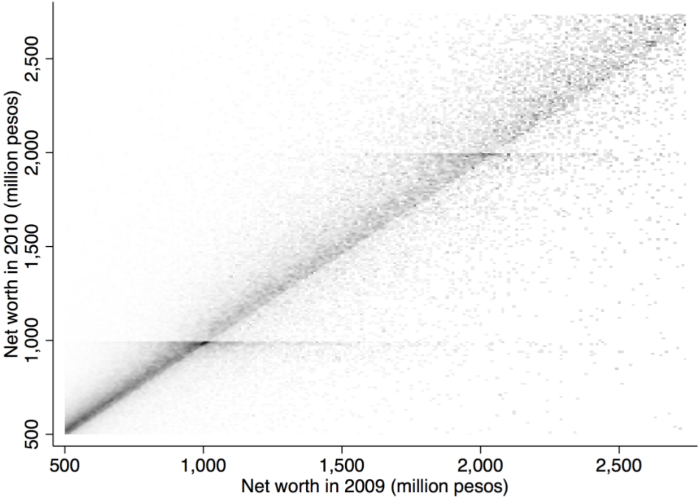
Thinking outside of the box to get new data



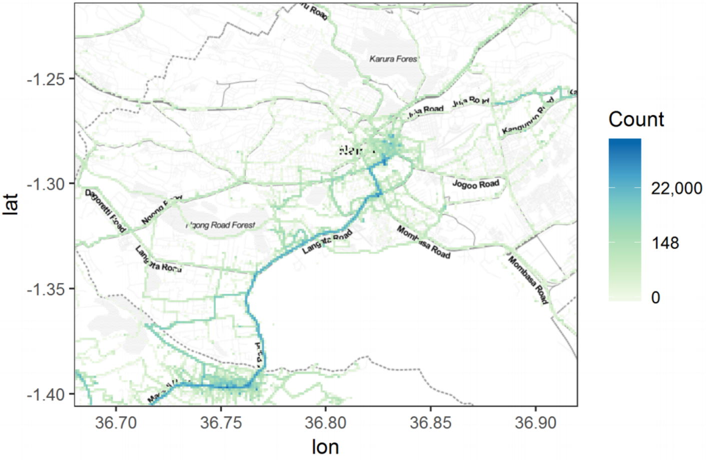
Thinking outside of the box to get new data



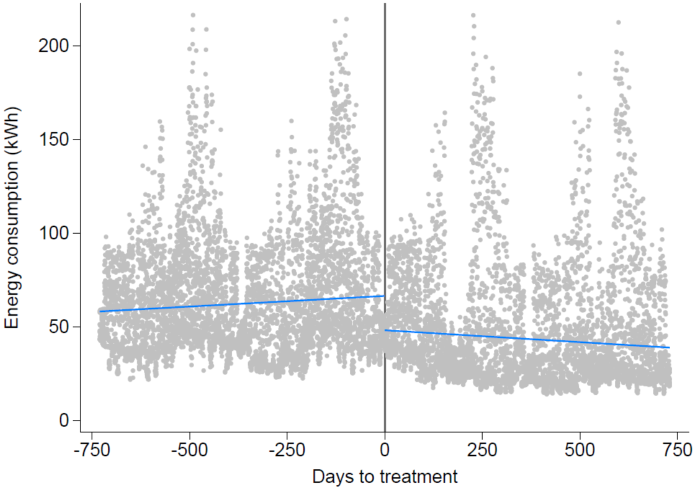
Thinking outside of the box to get new data



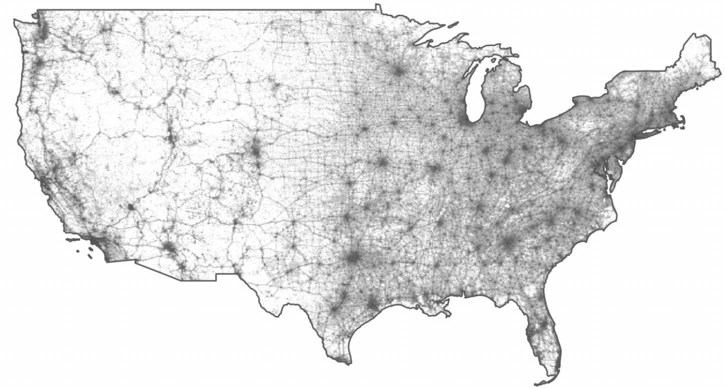
Thinking outside of the box to get new data



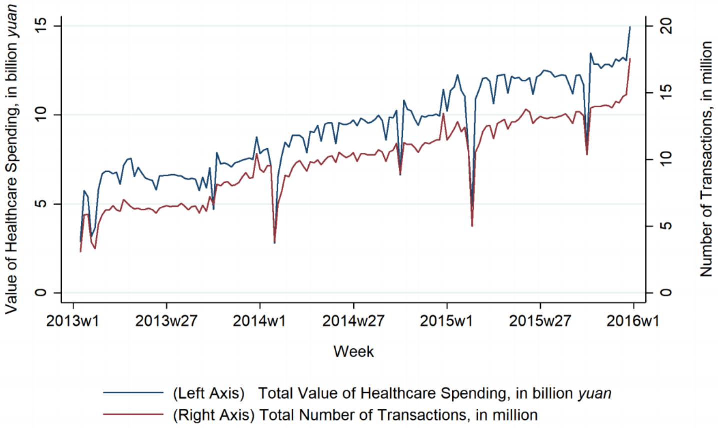
Thinking outside of the box to get new data



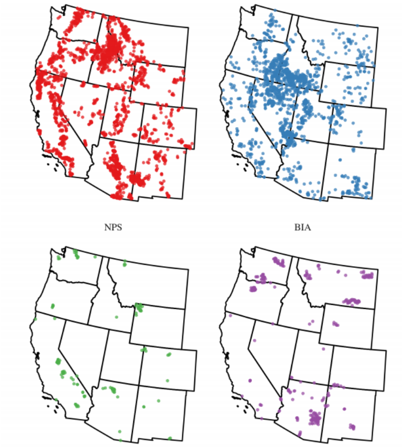
Thinking outside of the box to get new data



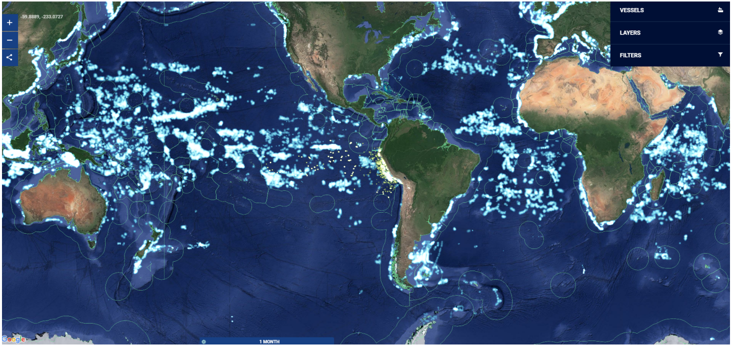
Thinking outside of the box to get new data



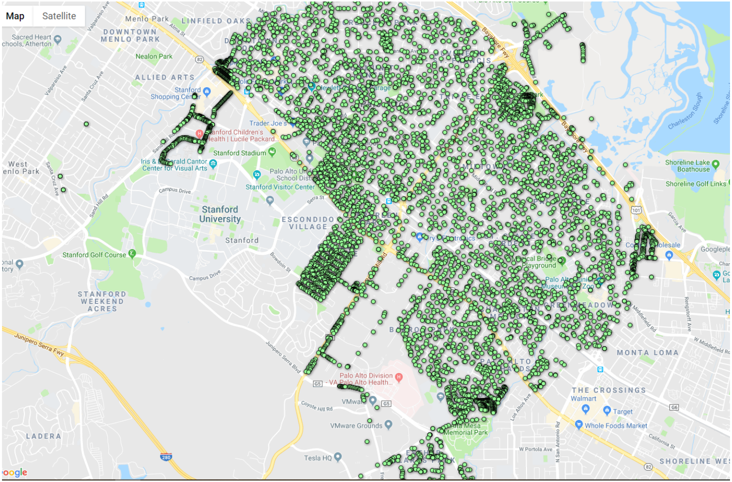
Thinking outside of the box to get new data



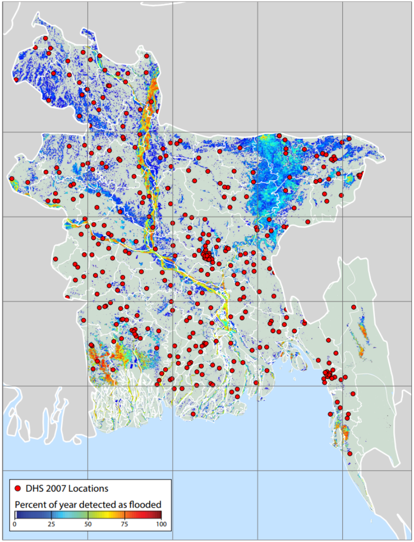
Thinking outside of the box to get new data



Thinking outside of the box to get new data



Thinking outside of the box to get new data



With Big Data come big responsibility

(Especially) with Big Data, we have to be careful:

- Just because a dataset is large doesn't mean it's unbiased
- Large data can also have errors
- And come with additional concerns too (privacy, etc)

With Big Data come big responsibility

(Especially) with Big Data, we have to be careful:

- Just because a dataset is large doesn't mean it's unbiased
- Large data can also have errors
- And come with additional concerns too (privacy, etc)

$\rightarrow$ **It's important to understand what we're using**

With Big Data come big responsibility

(Especially) with Big Data, we have to be careful:

- Just because a dataset is large doesn't mean it's unbiased
- Large data can also have errors
- And come with additional concerns too (privacy, etc)

→ **It's important to understand what we're using**

→ (The following slides owe credit to Tamma Carleton)

With Big Data come big responsibility

We typically interact with three types of data:

1. Raw, out of the source
2. Processed "in house"
3. Processed "out of the house"

→ All of these data can be used as Y, D, or X (or even Z)

→ Each has its own pros and cons

Raw data

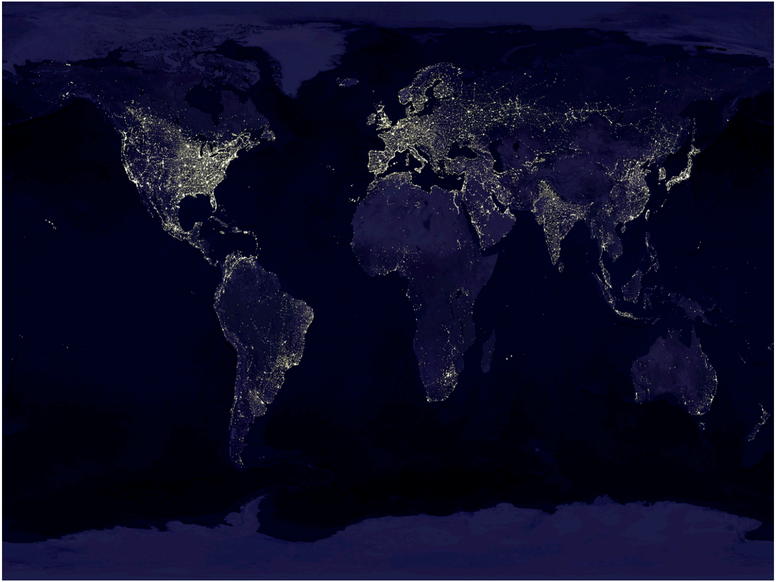
Major pros of raw data:

- We know what we're dealing with
- We get a fighting chance to understand measurement error and bias

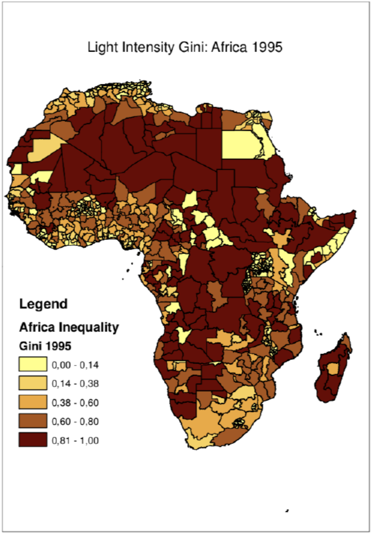
Major cons of raw data:

- Raw data are often not exactly what we want
- We have to be careful when we use them as a proxy

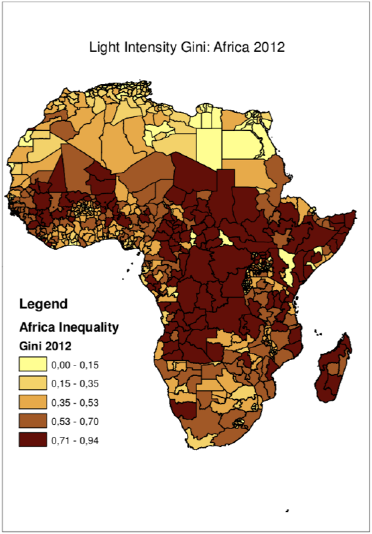
Raw data



Raw data

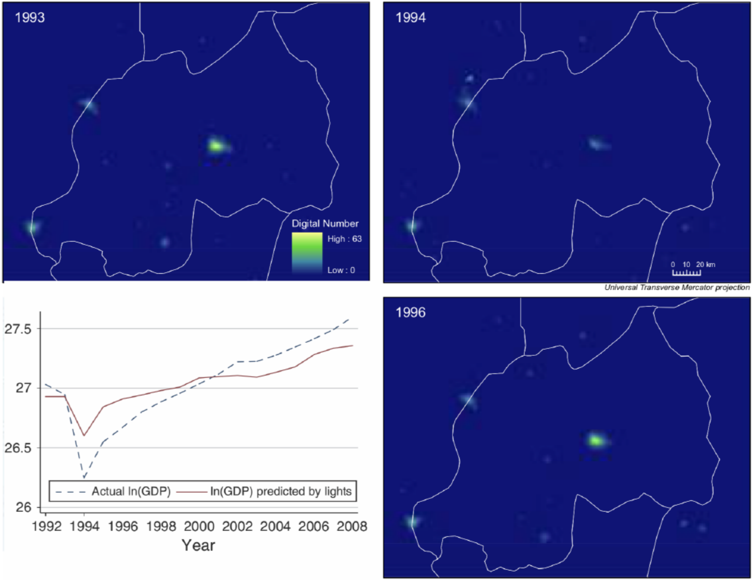


Light Intensity Gini: Africa 1995
Legend
Africa Inequality
Gini 1995
0,00 - 0,14
0,14 - 0,38
0,38 - 0,60
0,60 - 0,80
0,81 - 1,00

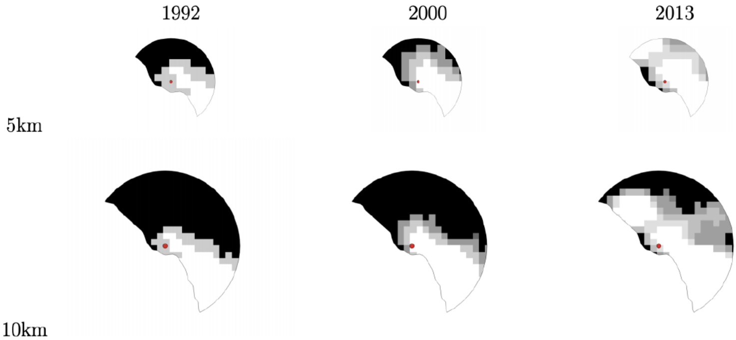


Light Intensity Gini: Africa 2012
Legend
Africa Inequality
Gini 2012
0,00 - 0,15
0,15 - 0,35
0,35 - 0,53
0,53 - 0,70
0,71 - 0,94

Raw data


1993
Digital Number
High : 63
Low : 0
1994
0 10 20 km
Universal Transverse Mercator projection
1996
27.5
27
26.5
26
1992 1994 1996 1998 2000 2002 2004 2006 2008
Year
Actual ln(GDP)
ln(GDP) predicted by lights

Raw data



Processed in house

Major pros of home-grown data:

- We know what we're dealing with
- We get a fighting chance to understand measurement error and bias

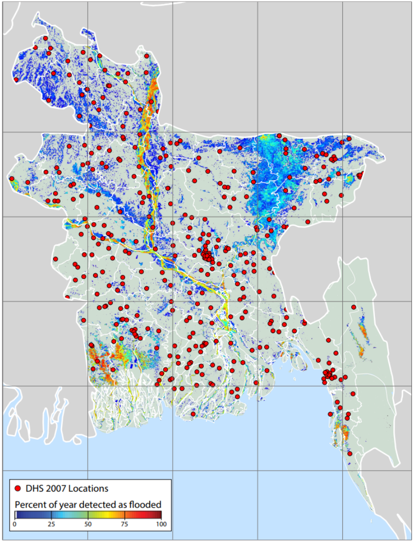
Major cons of home-grown data:

- This takes a lot of time and effort
- And we don't always have the right toolkit

Processed in house

Processed in house

Processed in house



Processed out of the house

Major pros of outsourced data:

- We leverage external expertise
- We potentially have less measurement error than the in-house version
- This is a lot less work than the

Major cons of outsourced data:

- We don't know exactly what we're measuring
- We can't look "under the hood" to uncover bias

Processed out of the house

Processed out of the house

Big Data wrap-up

Just like with small data, you need to know what you've got:

- Big Data allow for new possibilities
- But require additional processing tools and time
- A careful combination of in-house and out-of-house work can yield benefits

Estimation vs. prediction

This class has been about asking:

→ What is the causal effect of D on Y?

→ Aka, in

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

what is $\hat{\tau}$?

- Focus is on **unbiasedness**

Estimation vs. prediction

This class has been about asking:

→ What is the causal effect of D on Y?

→ Aka, in

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

what is $\hat{\tau}$?

- Focus is on **unbiasedness**

Machine learning instead asks:

→ What is the best guess of some outcome?

→ What is $\hat{Y}$?

- Want to consider a **bias-variance tradeoff**

Estimation vs. prediction

An estimation problem:

"What is the causal effect of my rain dance on rainfall today?"

$\rightarrow$ Estimation problem: rain dances (maybe) affect rainfall

Estimation vs. prediction

An estimation problem:

"What is the causal effect of my rain dance on rainfall today?"

$\rightarrow$ Estimation problem: rain dances (maybe) affect rainfall

A prediction problem:

"Do I need an umbrella today?"

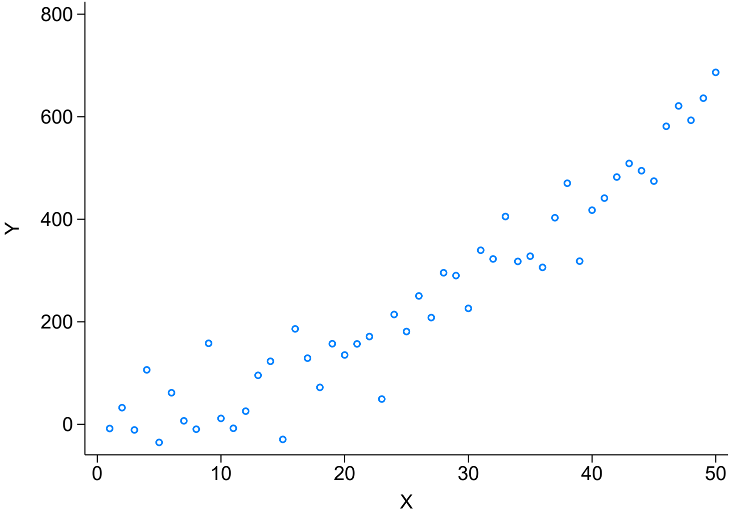
$\rightarrow$ Prediction problem: rainfall doesn't depend on umbrellas

Basics of machine learning

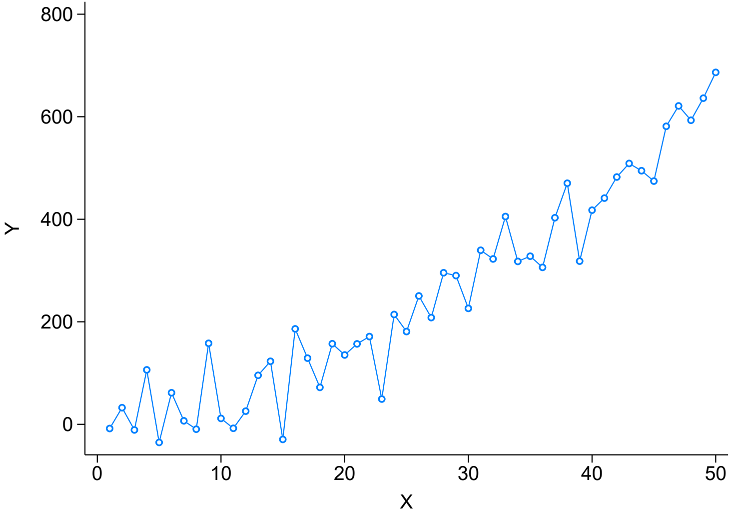
Machine learning is just methods trying to generate predicions:

- Given a dataset with outcome Y and covariates $\mathbf{X}$, what function $f(\mathbf{X})$ best predicts Y?
- Note the difference between this and causal inference!

Predicting $\hat{Y}$



Predicting $\hat{Y}$



Predicting $\hat{Y}$ without overfitting

We want to come up with a good estimate of $\hat{Y}$...
... But we need to watch out for overfitting!

Predicting $\hat{Y}$ without overfitting

We want to come up with a good estimate of $\hat{Y}$...
... But we need to watch out for overfitting!

Machine learning typically uses three steps:

1. **In-sample prediction**:
   - Use an algorithm to generate the best in-sample prediction
   - Many ways to do this, but imagine running thousands of OLS regressions

Predicting $\hat{Y}$ without overfitting

We want to come up with a good estimate of $\hat{Y}$...
... But we need to watch out for overfitting!

Machine learning typically uses three steps:

1. **In-sample prediction**:
   - Use an algorithm to generate the best in-sample prediction
   - Many ways to do this, but imagine running thousands of OLS regressions
2. **Cross-validation**:
   - Instead of using the whole sample for step (1)...
   - Split the sample into pieces...
   - Do step (1) on one part, and predict $\hat{Y}$ on the other part
   - Record how well the model fits (eg $Y - \hat{Y}$)

Predicting $\hat{Y}$ without overfitting

We want to come up with a good estimate of $\hat{Y}$...
... But we need to watch out for overfitting!

Machine learning typically uses three steps:

1. **In-sample prediction**:
   - Use an algorithm to generate the best in-sample prediction
   - Many ways to do this, but imagine running thousands of OLS regressions
2. **Cross-validation**:
   - Instead of using the whole sample for step (1)...
   - Split the sample into pieces...
   - Do step (1) on one part, and predict $\hat{Y}$ on the other part
   - Record how well the model fits (eg $Y - \hat{Y}$)
3. **Repeat**:
   - Do this several times over different sample splits
   - Pick the final model that does best

The in-sample prediction step

It's worth unpacking this a bit further:

- **Goal:** Produce the best guess at $\hat{f}(\mathbf{X})$
- This typically involves being very flexible: interactions between Xs
- We know we want to avoid over-fitting
- Just running a ton of OLS regressions is a slow way to do this

The in-sample prediction step

Function class $\mathcal{F}$ (and its parametrization)	*Regularizer $R(f)$*
Global/parametric predictors	
Linear $\beta'x$ (and generalizations)	Subset selection $\|\|\beta\|\|_0 = \sum_{j=1}^{k} 1_{\beta_j \neq 0}$
	LASSO $\|\|\beta\|\|_1 = \sum_{j=1}^{k} \|\beta_j\|$
	Ridge $\|\|\beta\|\|_2^2 = \sum_{j=1}^{k} \beta_j^2$
	Elastic net $\alpha\|\|\beta\|\|_1 + (1-\alpha)\|\|\beta\|\|_2^2$
Local/nonparametric predictors	
Decision/regression trees	Depth, number of nodes/leaves, minimal leaf size, information gain at splits
Random forest (linear combination of trees)	Number of trees, number of variables used in each tree, size of bootstrap sample, complexity of trees (see above)
Nearest neighbors	Number of neighbors
Kernel regression	Kernel bandwidth
Mixed predictors	
Deep learning, neural nets, convolutional neural networks	Number of levels, number of neurons per level, connectivity between neurons
Splines	Number of knots, order
Combined predictors	
Bagging: unweighted average of predictors from bootstrap draws	Number of draws, size of bootstrap samples (and individual regularization parameters)
Boosting: linear combination of predictions of residual	Learning rate, number of iterations (and individual regularization parameters)
Ensemble: weighted combination of different predictors	Ensemble weights (and individual regularization parameters)

A few cautions with ML models

ML is a great tool, but we have to be careful!

- **Do NOT try to interpret the function!**
- The ML model gives you $\hat{Y}$...but *not* $\hat{\tau}$!
- We're not recovering causal effects
- And we don't get standard errors
- And the models are typically unstable

A few cautions with ML models

ML is a great tool, but we have to be careful!

- **Do NOT try to interpret the function!**
- The ML model gives you $\hat{Y}$...but *not* $\hat{\tau}$!
- We're not recovering causal effects
- And we don't get standard errors
- And the models are typically unstable

Next time: ML meets causal inference

Recap

TL;DR:

1. New datasets open new questions
2. Machine learning offers opportunity
3. Both require some careful consideration or tweaks to be useful for us

For next class

Topics:

- Big data and machine learning II

Readings:

- Burlig et al (2019)